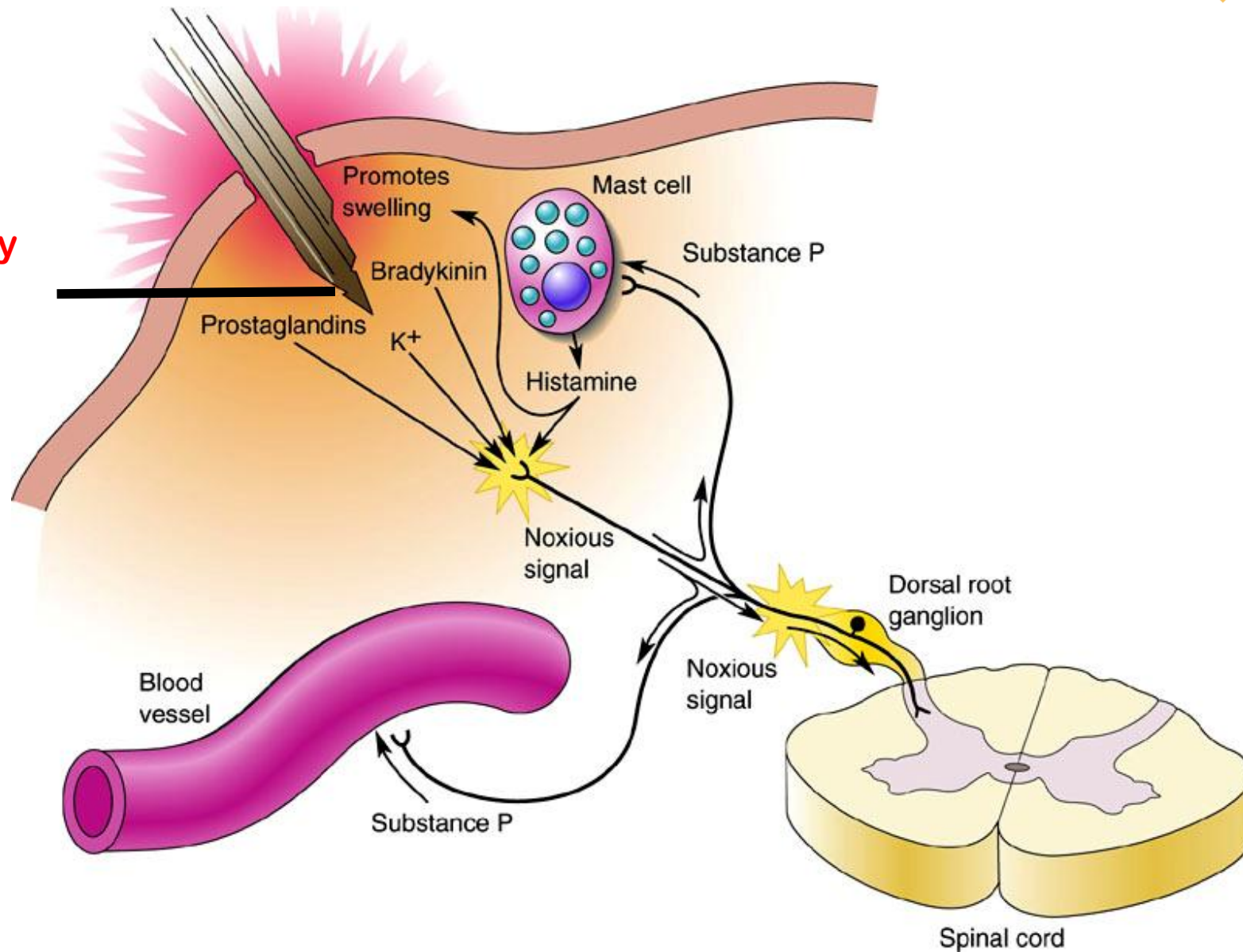


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Nonsteroidal Anti-Inflammatory Drugs (NSAIDs)

**Inflammatory
stimulus**



Peripheral mechanisms of pain



Inflammation

- is a normal, protective response to tissue injury caused by **physical trauma, noxious chemicals, or microbial agents**. However
- in certain condition the **immune response** can become inappropriate leading to a process of **chronic inflammation** like in **rheumatoid arthritis**.



Therapeutic strategies:

- **Treatment** of patients with **inflammation** involves **two primary goals**:
 - **Relief of pain**, which is often the presenting symptom and the major continuing complaint of the patient
 - **Slowing or arrest** of the **tissue damage**.





The drugs used in the treatment of inflammations include:

1.NSAIDs

2.Disease-modifying antirheumatic drugs(DMARDs)

3.Glucocorticoids

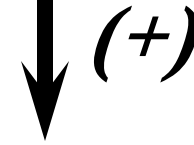




Eicosanoids

- Types:** Includes **prostanoids, leukotrienes, lipoxins, and epoxyeicosatrienoic acids (EETs).**
- Biosynthesis:** Eicosanoids are synthesized through three major enzyme pathways:
- Cyclooxygenase (COX) pathway → Produces prostanoids.**
- Lipoxygenase (LOX) pathway → Produces leukotrienes and lipoxins.**
- Cytochrome P450 pathway → Produces EETs.**

Inflammatory stimulus



Ex

Phospholipids

Phospholipase A₂



In



Arachidonic acid

Cyclooxygenase (COX)

Endoperoxides

PGs

TxA₂

5-lipoxygenase

Leucotrienes

15-lipoxygenase

Lipoxins



Prostanoids (Subset of Eicosanoids)

•**Definition:** A subclass of eicosanoids produced exclusively via the **cyclooxygenase (COX) pathway**.

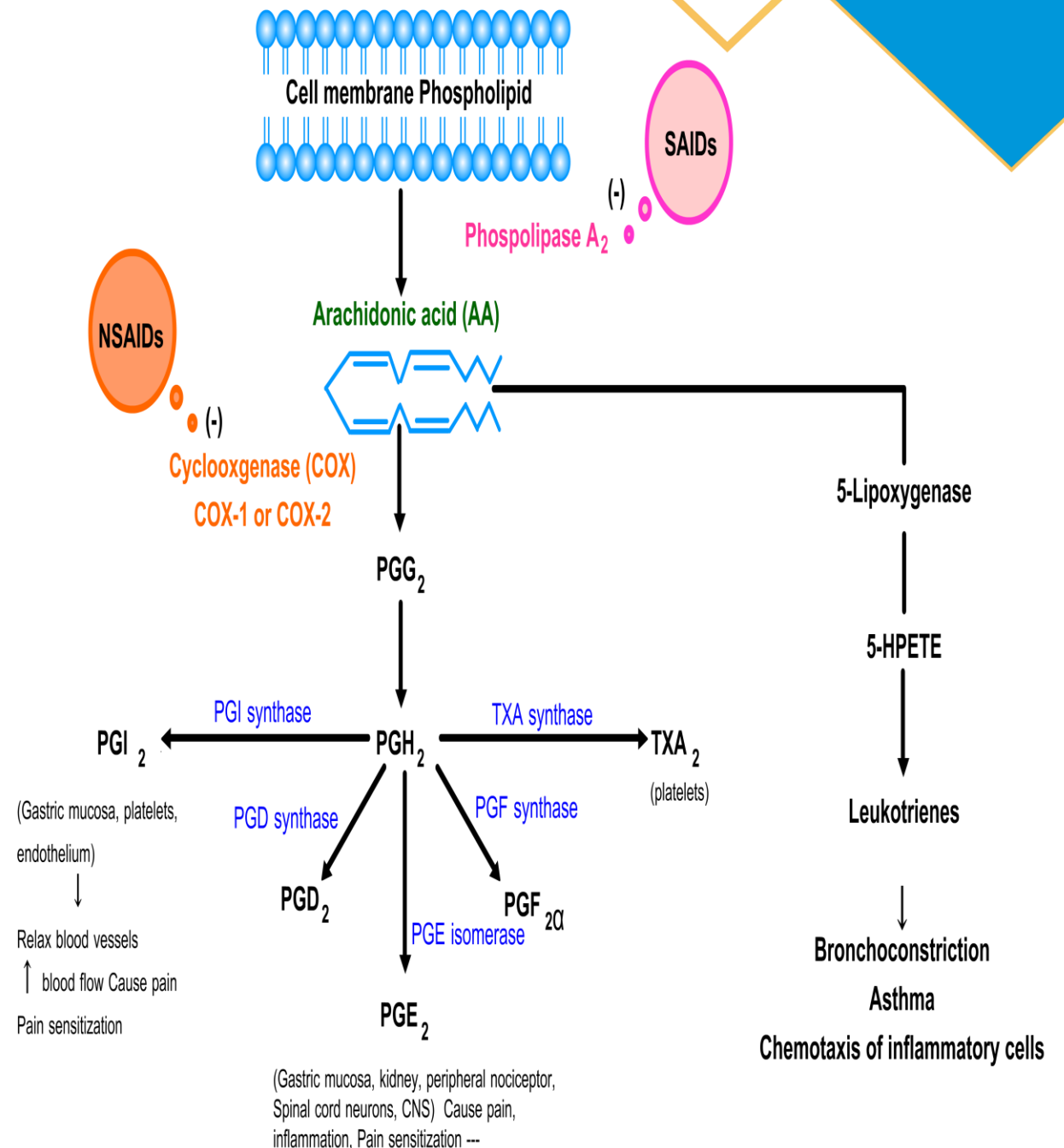
•**Types:** Includes **prostaglandins (PGs)**, **prostacyclin (PGI₂)**, and **thromboxanes (TXs)**.

•**Biosynthesis:** Arachidonic acid is converted by COX enzymes into prostaglandin H₂ (PGH₂), which is further metabolized into:

- **Prostaglandins (PGE₂, PGD₂, PGF₂α, PGI₂)**
- **Thromboxanes (TXA₂, TXB₂)**

•**Functions:**

- **Prostaglandins (PGs):** Mediate inflammation, fever, pain, and smooth muscle contraction.
- **Prostacyclin (PGI₂):** Inhibits platelet aggregation and promotes vasodilation.
- **Thromboxanes (TXs):** Promote platelet aggregation and vasoconstriction.



GENERAL FEATURES OF NSAIDs:

NSAIDs have *anti-inflammatory, antipyretic and analgesic* effects in different proportions. Non-selective agents, also *inhibit platelet aggregation*

- All NSAIDs are **weak organic acids** except, *nabumetone*.
- Most are **well absorbed orally**.
- Most are **highly protein-bound (98%)**
- Most are **highly metabolized in liver** and **excretion via kidney** with **variable enterohepatic circulation**.

PROSTANOIDS (PGs & Txs)

PGI₂ (prostacyclin) is located predominantly in vascular endothelium.

Main effects:

- vasodilatation
- inhibition of platelet aggregation

TxA₂ is found in the platelets.

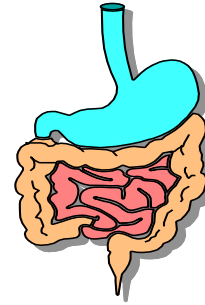
Main effects:

- platelet aggregation
- vasoconstriction



PGE2 – main effects:

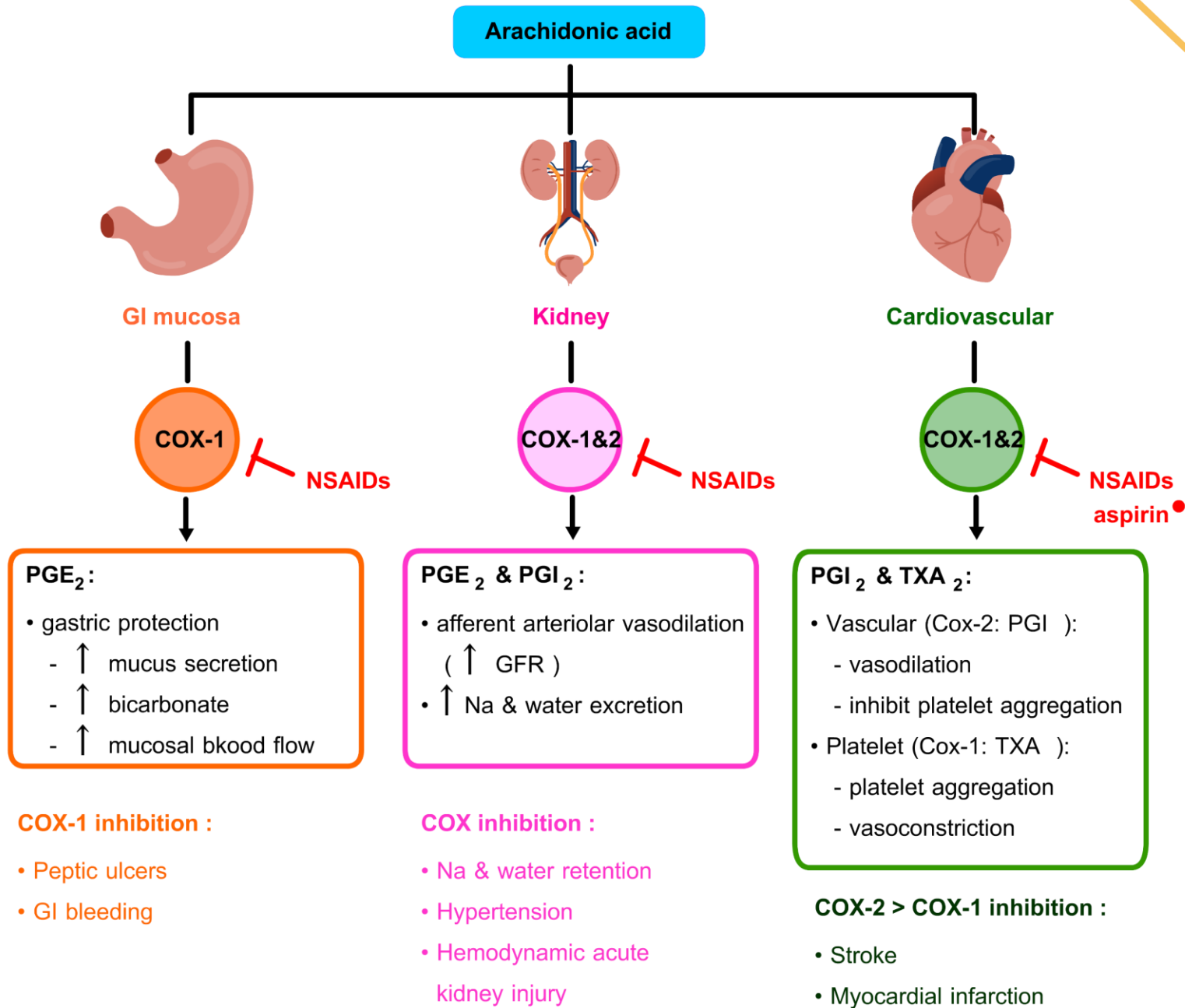
- Decrease of gastric acid secretion
- Contraction of pregnant uterus.
- Contraction of GI smooth muscle



PGF2 α – main effects:

- Contraction of bronchi 
- Contraction of myometrium





• Low dose aspirin irreversibly inhibits platelet COX-1





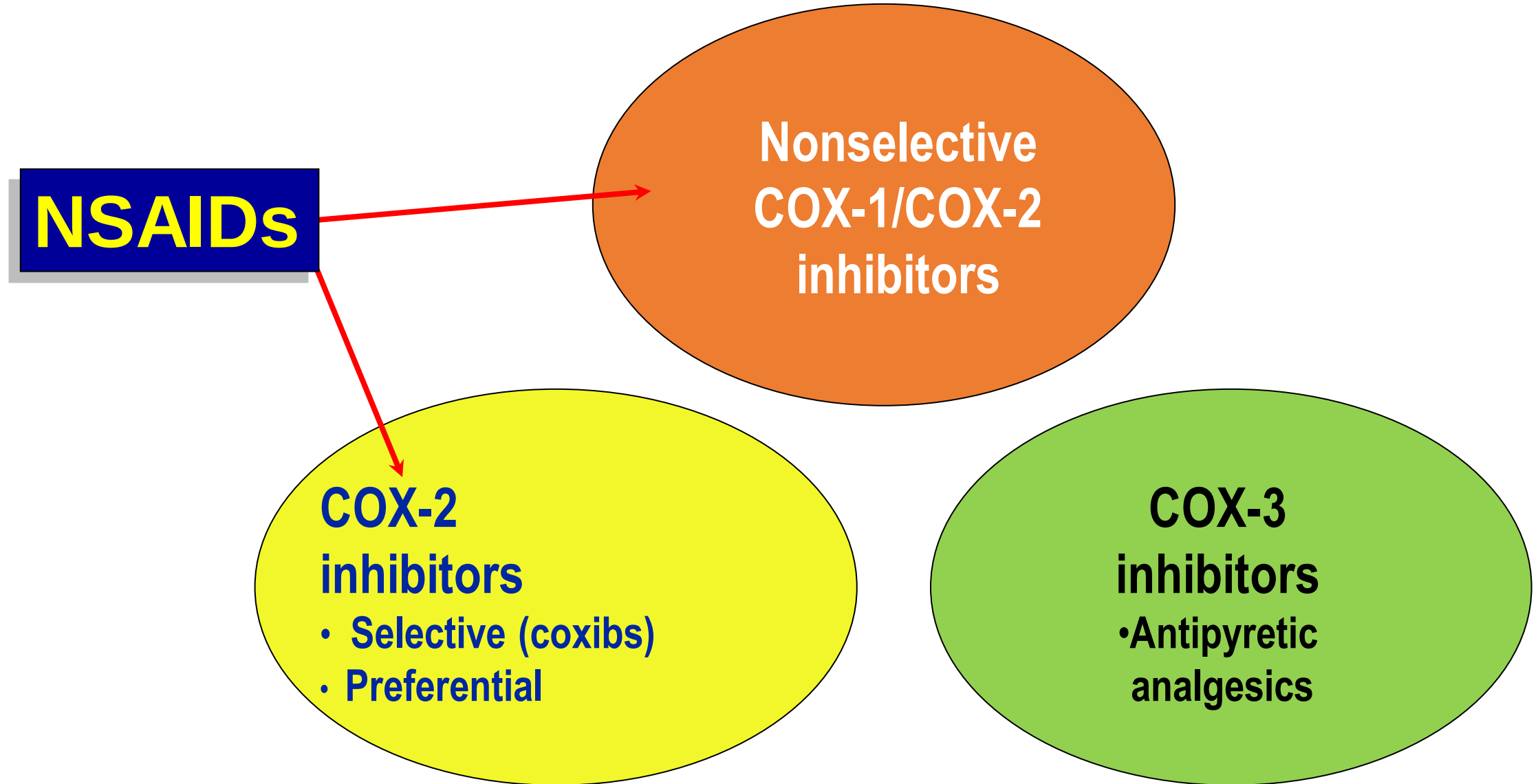
Cyclooxygenase (COX) is found bound to the endoplasmatic reticulum.

It exists in 3 isoforms:

- **COX-1** (constitutive) acts in physiological conditions.
- **COX-2** (inducible) is induced in inflammatory cells by pathological stimulus.
- **COX-3** (in brain).



COX inhibitors



Nonselective COX-1/COX-2 inhibitors (Classical NSAIDs)

1. Salicylates

- **Aspirin** (Acetylsalicylic acid): Used as an anti-inflammatory, pain reliever, and for reducing fever.
- **Salsalate**: Used to treat pain and inflammation in conditions such as arthritis.

2. Phenylacetates

- **Diclofenac** (often as sodium or potassium salt): A nonsteroidal anti-inflammatory drug (NSAID) used to treat pain and inflammation.
- **Bendazac**: Used in the treatment of cataracts.

3. Indolacetates

- **Indomethacin**: A strong NSAID commonly used to reduce fever, pain, stiffness, and swelling.

4. Enolates

- **Allopurinol** (technically an isomer of hypoxanthine): Used to lower uric acid levels in people with gout and kidney stones.

5. Fenamates

- **Mefenamic Acid**: Used for treating mild to moderate pain, including menstrual pain.
- **Tolfenamic Acid**: Commonly used in veterinary medicine to reduce inflammation and pain.

6. Propionates

- **Ibuprofen** (technically an isobutylphenyl propionic acid): Widely used as an anti-inflammatory and pain reliever.
- **Naproxen**: Similar to ibuprofen, used for pain relief, especially in conditions like arthritis.

7. Butylpyrazolidindiones

- **Phenylbutazone**: Often used in veterinary medicine for its anti-inflammatory and pain relief properties.

8. Pyrazolones

- **Phenazone** (Antipyrine): Used to relieve pain and fever.
- **Metamizole** (Dipyrone): A powerful analgesic and antipyretic.



General Uses of NSAIDs:

- 1. Painful conditions:*** for mild-moderate pain like toothache, Inflammatory arthropathy (e.g., rheumatoid arthritis) , dysmenorrhea (menstrual pain), gout, metastatic bone pain, headache, migraine, postoperative pain and renal colic.
- 2. Pyrexia (fever).***
- 3. Inhibition of platelet aggregation:*** thrombotic conditions (like CVA, MI and DVT)
- 4. They are also given to neonate infants whose ductus arteriosus is not closed (patent ductus arteriosus) within 24 hours of birth.***



- ***Adverse effects of the NSAIDs:***

1. **CNS:** Tinnitus, and dizziness, confusion
2. **CVS:** Fluid retention, hypertension, edema
3. **GIT:** NSAIDs are all gastric irritants and can be associated with GI ulcers and bleeding (this effect is less with selective COX II inhibitors)
4. **Hematologic:** Rarely cause thrombocytopenia
5. **Hepatic:** Abnormal liver function tests
6. **Pulmonary:** Bronchospasm (increase asthma symptoms)
7. **Rashes and pruritus** may be caused by any type of NSAIDs.
8. **Renal:** Renal insufficiency, and some time renal failure,



Types of NSAIDs:

- **1- Non selective:** inhibit both COX-I and COX-II e.g. *aspirin*.
- **2- Selective COX-II Inhibitors:** inhibit COX II, less gastric side effects with nearly similar effectiveness; do not affect platelet function at usual doses. They may increase the incidence of edema and hypertension e.g. *celecoxib*.
- **3- COX III inhibitor:** act centrally (e.g. *acetaminophen*).



Types of NSAIDs:

Aspirin (Acetyl Salicylic Acid)

- Aspirin (**prototype NSAID agent**) is unique among the NSAIDs in that it *non selective irreversibly inactivates cyclooxygenase*, (**others are reversible**) it's well absorbed orally.
- About 50–80% of salicylate in the blood is bound by protein.
- About **80% of therapeutic dose** of salicylic acid is **metabolized in the liver**. These metabolic pathways have only a **limited capacity**. With **large doses**, the kinetics switch from first order to **zero order**, as metabolic pathway become saturated, renal excretion becomes increasingly important.
- **Alkalinization of urine increases the rate of renal excretion** (so it is important in case of salicylate intoxication).

Acetaminophen (Paracetamol)

It inhibits prostaglandin synthesis in the CNS (inhibit COX-3). This explains its *antipyretic* and *analgesic properties* with weak anti-inflammatory activity. It is drug of choice for children.

Acetaminophen **does not affect platelet function** or increase bleeding time. It is **well absorbed orally**, with **less GIT side effects**.

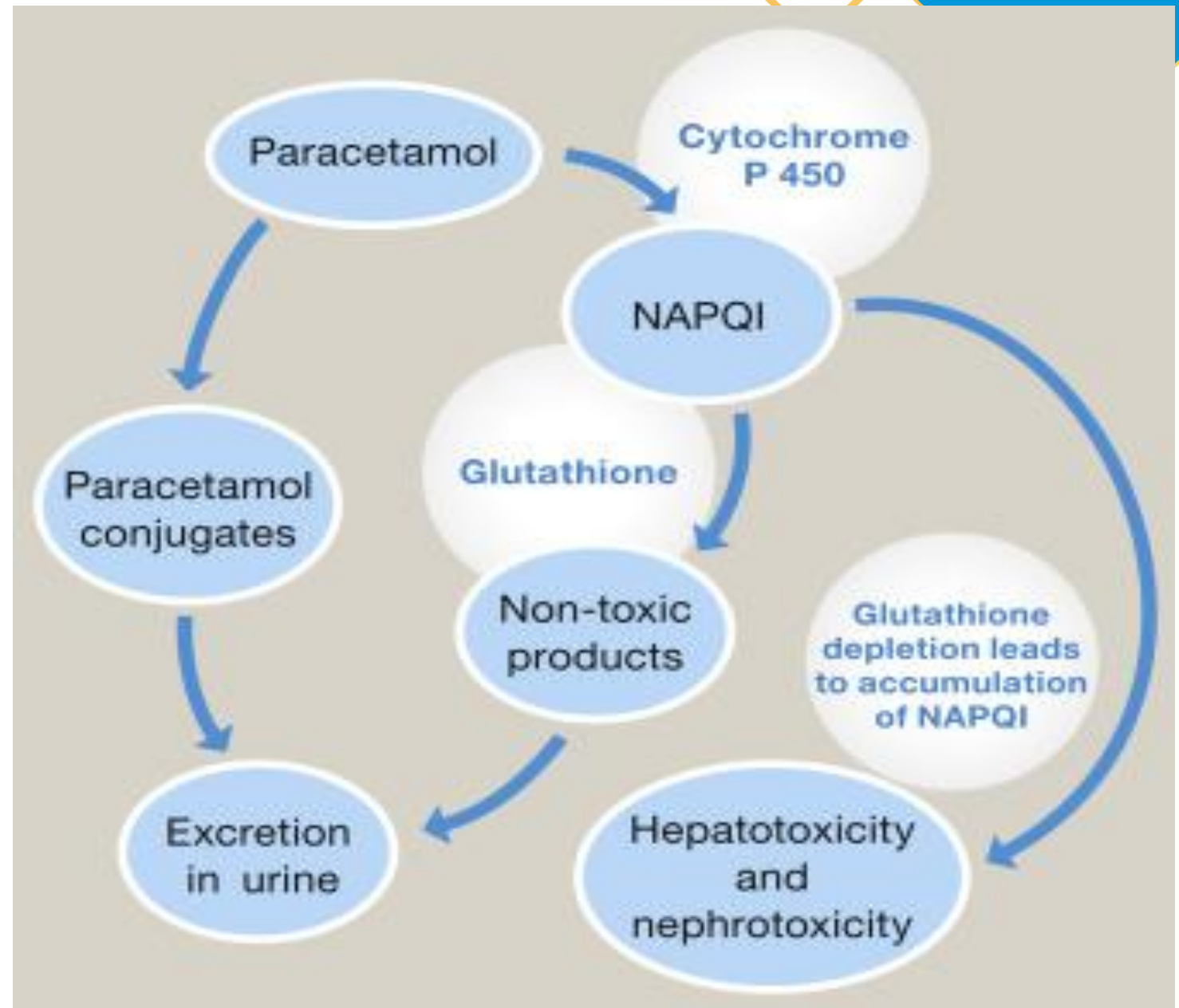
It metabolized in the liver by:

1. **Glucuronidation** is account for 40- 65% of the metabolism.
2. **Sulfation** (sulfate conjugation) may account for 20–40%.
3. The hepatic **cytochrome P450** system 5%.

Clinical use: Analgesic & Antipyretic

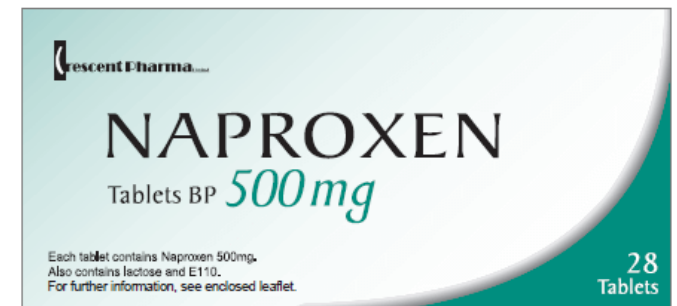
Toxicity:

- **hepatotoxicity:** it is from one of its metabolites, *N-acetyl-p-benzoquinoneimine* (NAPQI) **produced by cytochrome P450 system** in the liver.
- Paracetamol overdose leads to the accumulation of NAPQI, which undergoes conjugation with glutathione a natural antioxidant and detoxified
- Following overdose, detoxification pathway becomes saturated leading to acute toxicity (**Hepatic necrosis**) which is very serious and potentially **life-threatening condition** that can be prevented by administration of **paracetamol antidote** (N-Acetyl cysteine).



Propionic acid derivatives:

- Include: **Ibuprofen**, **naproxen**, **fenoprofen**, **ketoprofen**, and **oxaprozin**. Their **GI side effects** are generally **less intense** than those of aspirin. **Oxaprozin** has the **longest half-life** and is administered **once daily**.



Acetic acid derivatives:

- Include: indomethacin, sulindac, and etodolac.
- **Sulindac** is an inactive **prodrug** that is closely related to indomethacin, *it suppresses familial intestinal polyposis and it may inhibit the development of colon cancer.* **Pancreatitis** is another side effect for indomethacin.



Oxicam derivatives:

- Include: **Tenoxicam**, **piroxicam** and **meloxicam**. They have **long half-lives**, which permit **once-daily** administration, *Meloxicam* is *preferentially inhibits COX-2 over COX-1*, particularly at **lower therapeutic dose**.



Heteroaryl acetic acids:

- Include: **Diclofenac** and **tolmetin**. **Diclofenac** is **more potent than indomethacin or naproxen**. An **ophthalmic preparation** is also available.



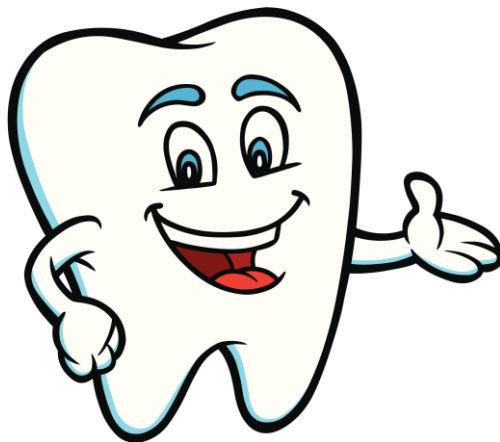
- **Ketorolac:**

- It use **mainly as an analgesic, not as an anti-inflammatory** drug. Used in the **treatment of postoperative pain & topical use for painful allergic conjunctivitis**



Fenamates:

- Include: **Mefenamic acid**, **flufenamic**, and **meclofenamate**.
- They can be **used for irregular vaginal bleeding and dysmenorrhea**.



- Celecoxib: It is selective for inhibition of COX-2 than of COX-1.

- Nimesulide It is relatively COX-2 selective.

Has a risk of hepatotoxicity.



- Licofelone It acts by inhibiting LOX (lipoxygenase) & COX.
- Nabumetone It is the only non-acid NSAID in current use.





**THANK YOU FOR YOUR
ATTENTION**

WILLIAM